



Blaze2D
A 2D Maxwell Solver in Rust

TECHNICAL REPORT

Blaze2D: A High-Performance Solver for Photonic Crystals

Achieving order-of-magnitude speedups through a mixed-precision LOBPCG algorithm and cache-aware architecture.

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Repository:
github.com/RnLe/blaze2d
Package:
pypi.org/project/blaze2d

Published
January 8, 2026

Imprint

Project: Blaze2D Technical Report
Title: Blaze2D: A High-Performance Solver for Photonic Crystals
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Date: January 8, 2026
Keywords: photonic crystal, plane-wave expansion, LOBPCG, mixed precision, MPB, benchmarks, Rust

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Package: <https://pypi.org/project/blaze2d/>

Source: <https://rnle.github.io/blaze2d/blaze/>

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Abstract

Photonic band structure calculations commonly rely on the Plane Wave Expansion (PWE) method, and MIT Photonic Bands (MPB) has long served as a widely used reference implementation for it [1,2]. MPB is trusted for its accuracy and is therefore a natural baseline for performance comparisons.

Blaze uses LOBPCG and mixed-precision arithmetic [3,4]. This report presents historical measurements. The archived datasets do not consistently record hardware identifiers or solver revisions. Series 4 and 5 use MPB tolerance $1e-7$ and Blaze tolerance $1e-4$; those comparisons are not at equal tolerance. Reported ratios apply to the recorded experiments.

1 Single-Core Performance

Performance benchmarks utilize the canonical square and hexagonal lattice configurations from Joannopoulos’ seminal 1997 Nature paper [5]. Unless otherwise specified, all data reflects the square lattice configuration at a resolution of 64, computing the lowest 8 bands with 20 k -points per segment between two high-symmetry points.

The computational cost of PWE solvers is dominated by Fast Fourier Transforms (FFTs) [1]. In this plane-wave formulation, Transverse Electric (TE) modes are more expensive to solve than Transverse Magnetic (TM) modes, because each operator application requires six FFTs rather than two for TM.

In these recorded single-core workloads, the f64 and mixed-precision Blaze runs are faster than the MPB runs. The mixed-precision measurements give an average speed ratio of about 2.9 $\times$. This ratio depends on the geometry, numerical settings, and execution environment.

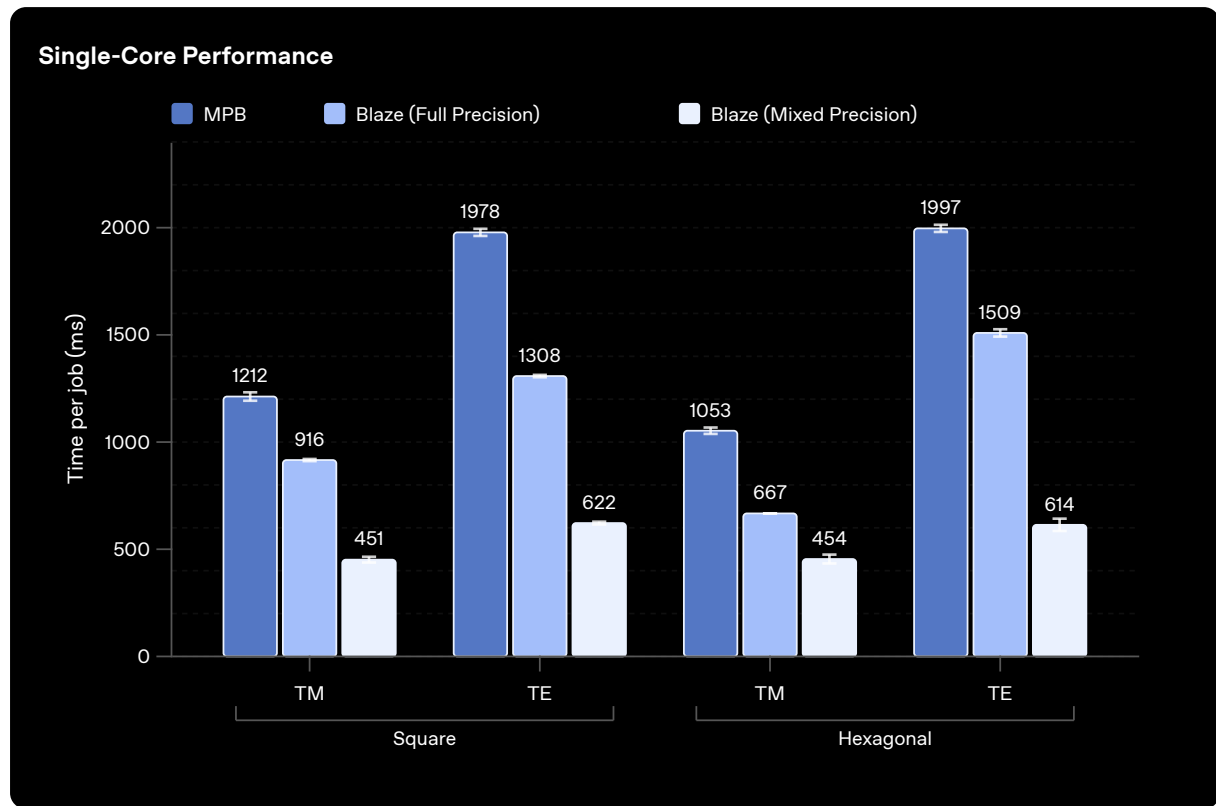


Figure 1: Blaze achieves 2.9 $\times$ average speedup over MPB on single-core workloads. Mixed precision (f32/f64) shown alongside full precision (f64).

2 Multi-Core Performance

The benchmark compares MPB threading within individual solves with Blaze scheduling independent configurations. The cost of synchronization and data movement can limit scaling for small problems.

Blaze schedules independent configurations across workers. In these measurements, full precision TE solves take approximately 240–350 ms and mixed precision runs cluster near 160 ms. The timing data alone do not identify a bandwidth limit or separate allocation, synchronization, and arithmetic costs [6].

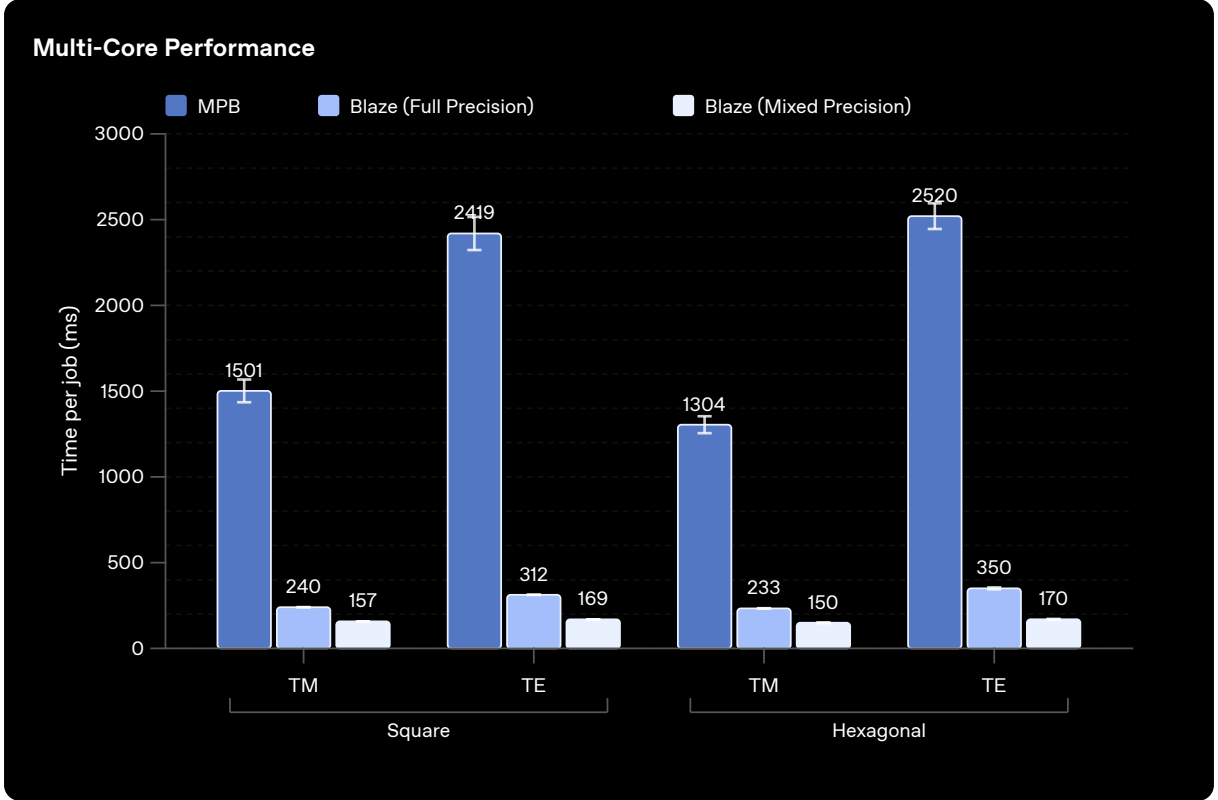


Figure 2: The recorded 16-thread workloads have an average MPB-to-Blaze timing ratio of 11.9. Mixed precision (f32/f64) shown alongside full precision (f64).

All subsequent benchmarks for Blaze are performed in Mixed Precision mode, unless otherwise specified.

3 Memory Efficiency

The recorded process-memory measurements are approximately 190 MB for MPB over much of this sweep. Process memory includes runtime and library allocations, so this observation does not establish that MPB reserves a fixed solver workspace.

Blaze uses less measured peak memory in these runs. A smaller working set can reduce data movement, but cache residency requires direct profiling and cannot be inferred from the memory ratio alone.

For a square grid with side length N , field storage grows as N^2 per band. Eigensolver workspaces also depend on band count, precision, and retained outputs. Process-level measurements may conceal that growth over a limited range.

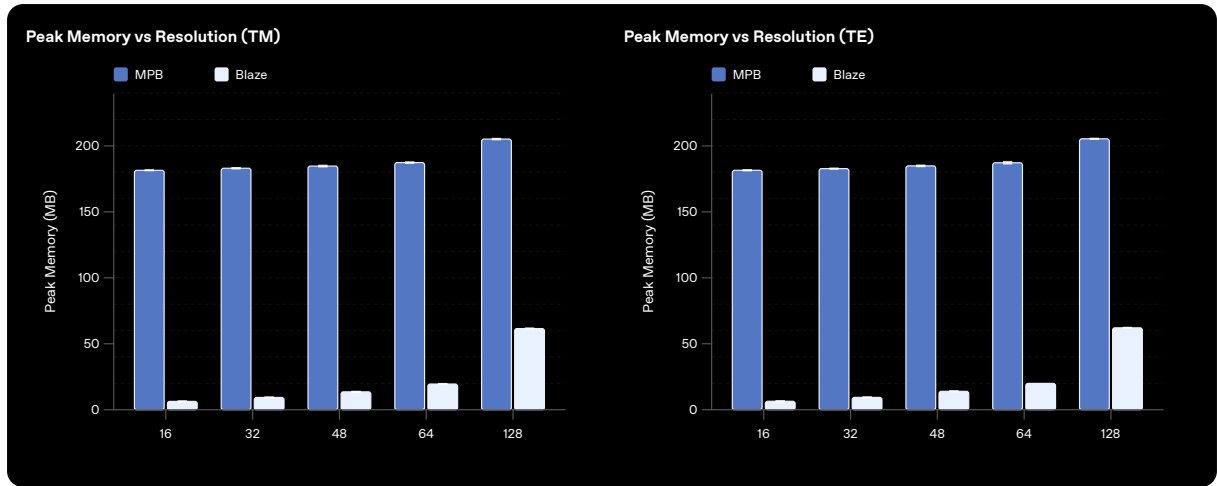


Figure 3: Peak memory against grid resolution, for both polarizations. MPB holds a fixed footprint of roughly 190 MB at every resolution, while Blaze allocates what the problem actually requires.

Both recorded process-memory curves remain nearly flat over the tested band-count range. This does not imply that storing more eigenvectors has no memory cost.

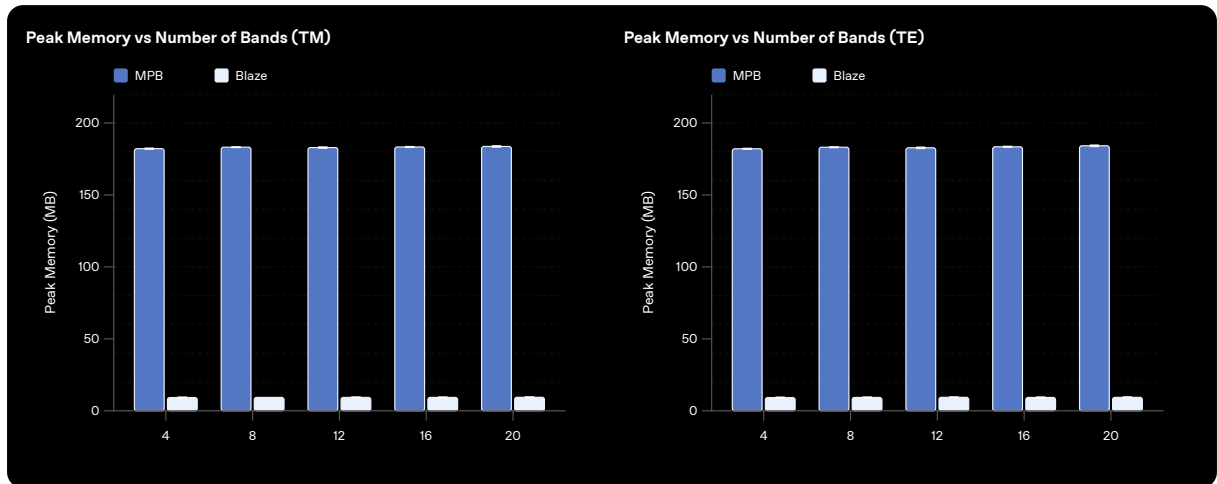


Figure 4: Peak memory against the number of requested bands. Neither solver's footprint grows with the size of the LOBPCG search block.

3.1 Memory Scaling Laws

The measured MPB-to-Blaze memory ratio is about 20× at the smallest grids and decreases as the grid grows. These finite-range measurements do not determine an asymptotic memory ratio.

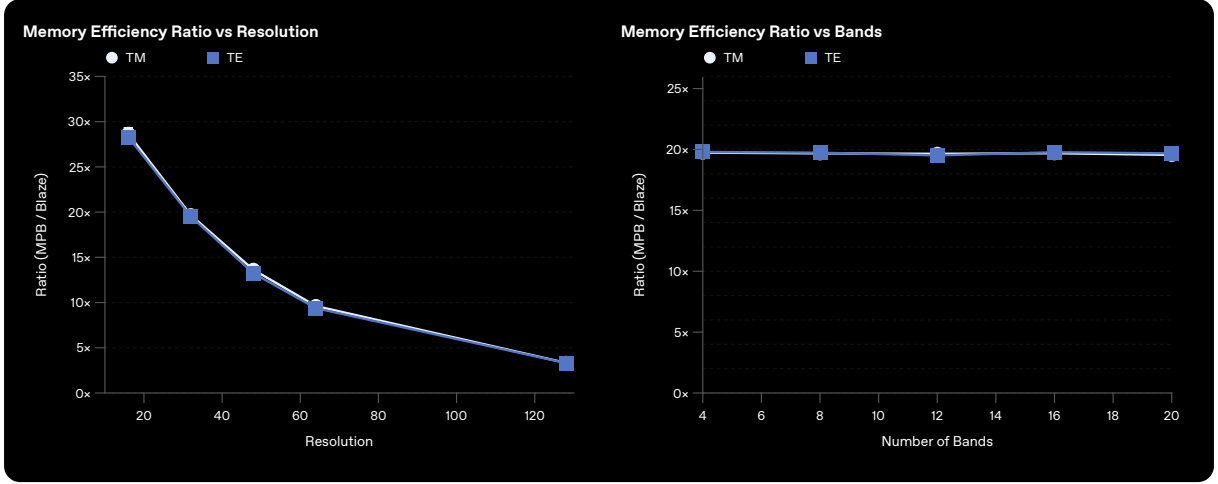


Figure 5: Ratio of MPB's peak memory to Blaze's, swept over resolution (left) and band count (right). The advantage is largest where MPB's fixed allocation dominates, and decays as the grid itself becomes the cost.

The fitted exponents over this finite range are 0.06 for MPB and 1.09 for Blaze. These are empirical process-memory trends, not asymptotic storage laws or proof of a particular allocation strategy.

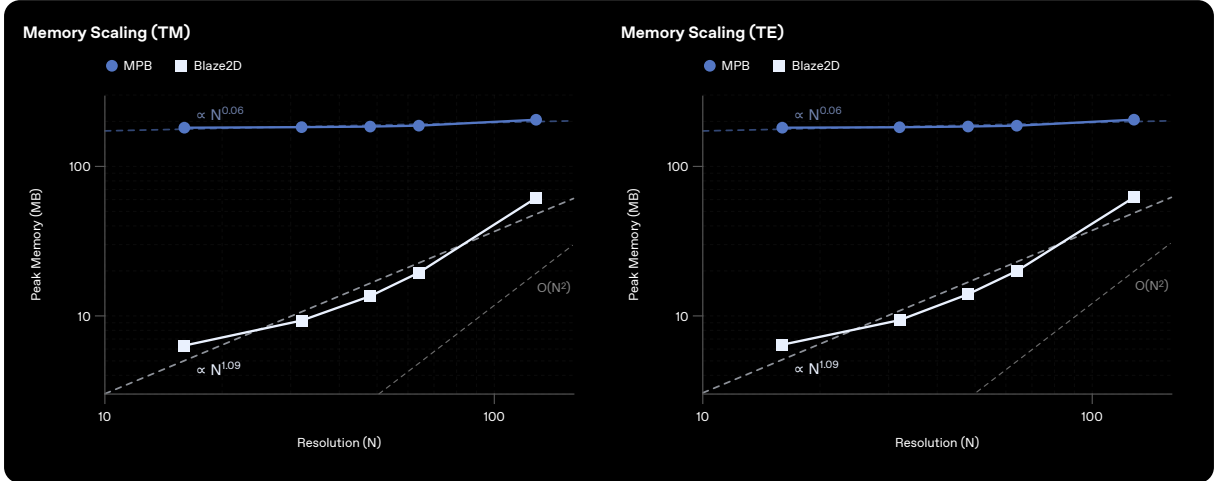


Figure 6: Peak memory against resolution on log-log axes, with power-law fits. MPB is flat at $N^{0.06}$, Blaze follows $N^{1.09}$, well below the $O(N^2)$ reference.

4 Resolution and Geometry

Grid resolution gives the number of samples along each unit-cell direction. Figure 7 shows the sampled permittivity of a circular rod. Boundary cells use subpixel averaging, related to the interface treatment described by [7]. This does not establish identical rasterization behavior to MPB. The required resolution depends on geometry, dielectric contrast, frequency, and the target observable. Check convergence under grid refinement.

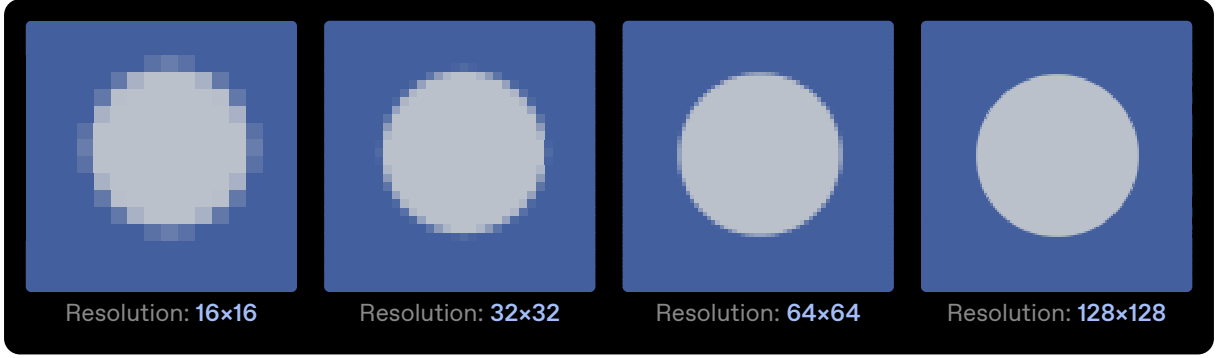


Figure 7: Smoothed permittivity map of a circular rod at four grid resolutions. The boundary cells take intermediate values from subpixel smoothing rather than snapping to $\varepsilon = 1$ or $\varepsilon = 13$, which is what keeps a coarse grid usable. On the website this figure is an interactive slider.

5 Accuracy Validation

The reduction in precision and memory footprint raises a direct question: do we gain speed at the cost of accuracy? To answer this question, we compared the eigenfrequencies from Blaze against a high-precision MPB reference along the full $\Gamma \rightarrow X \rightarrow M \rightarrow \Gamma$ path. In Figure 8, the MPB bands are drawn as lines and the Blaze eigenvalues as markers.

The comparison computes 20 bands and displays the lowest 10. Frequencies are compared as sorted eigenvalue sets at each wavevector, avoiding sensitivity to band labels near crossings and the upper edge of the solved window. This historical plotting convention differs from the current API, which preserves the tracked path returned by the solver.

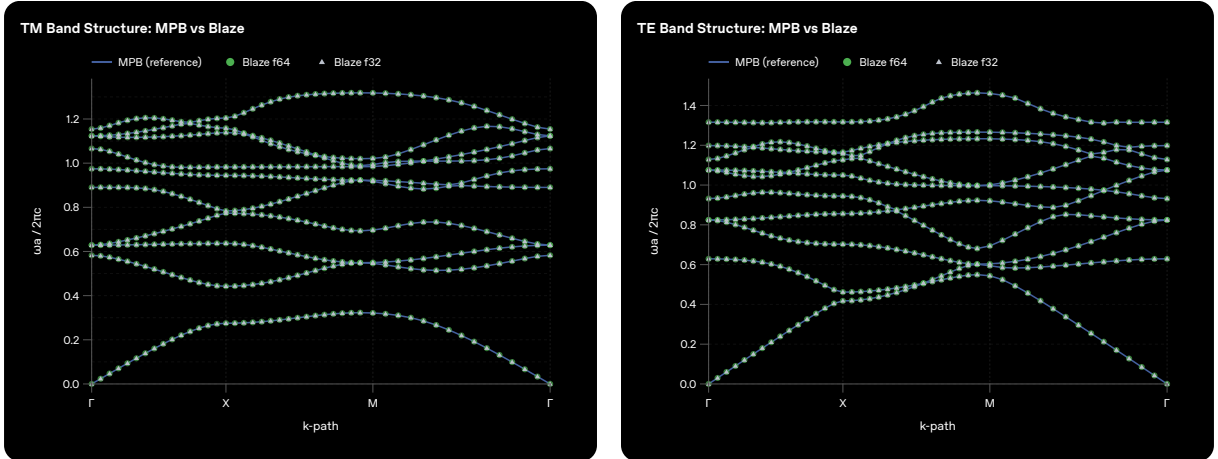


Figure 8: Band structures along $\Gamma \rightarrow X \rightarrow M \rightarrow \Gamma$ for TM (left) and TE (right). MPB is drawn as lines, Blaze in full and mixed precision as markers. The three curves are indistinguishable at this scale.

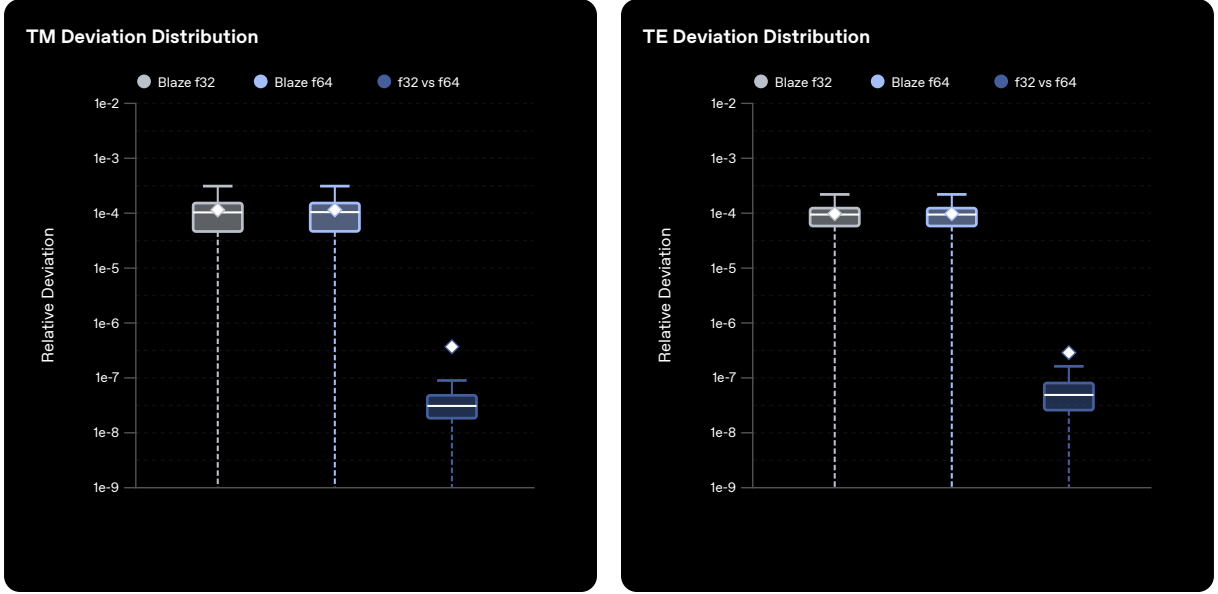


Figure 9: Distribution of the relative deviation across all bands and k -points, for TM (left) and TE (right). The first two boxes measure each Blaze run against MPB; the third measures the two Blaze runs against each other.

6 Deviation Analysis

The recorded median relative frequency difference from MPB is about 1×10^{-4} . Maxima are approximately 4×10^{-4} for TM and 3×10^{-4} for TE. These are differences between numerical runs, not certified discretization-error bounds.

The f32 and f64 Blaze runs differ by a median of about 1×10^{-7} and a maximum below 5×10^{-5} . Their agreement in these fixtures is consistent with mixed-precision LOBPCG studies using double-precision critical reductions [4]. It does not establish equal accuracy for other spectra, operator derivatives, or discretizations. Timing and memory ratios come from separate experiments and must be interpreted with their respective settings.

7 Parallel Scaling

The two solvers parallelize in different ways. MPB splits the work inside a single solve across threads, while Blaze runs many independent solves at the same time, one per thread, which suits the parameter sweeps it is built for. Figure 10 sweeps the thread count from 1 to 16 and measures throughput in solves per second, for a small problem ($N = 16$) and a large one ($N = 128$).

For the recorded $N = 16$ problem, Blaze throughput increases from about 28 to 220 solves per second across 16 threads. MPB threaded throughput stays near 14 solves per second. Separate MPB processes scale better than its threaded mode in this experiment.

For $N = 128$, Blaze throughput increases by about $3\times$ across the thread-count sweep. This supports scheduling independent configurations for these workloads; it does not establish a general limit on MPB parallelism.

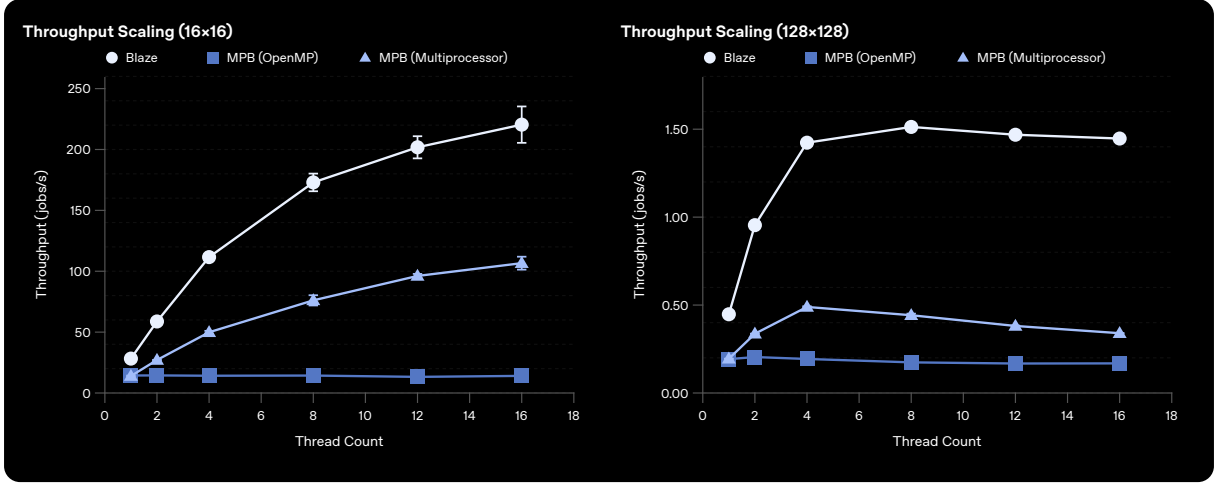


Figure 10: Throughput in solves per second against thread count, for a small problem ($N = 16$, left) and a large one ($N = 128$, right).

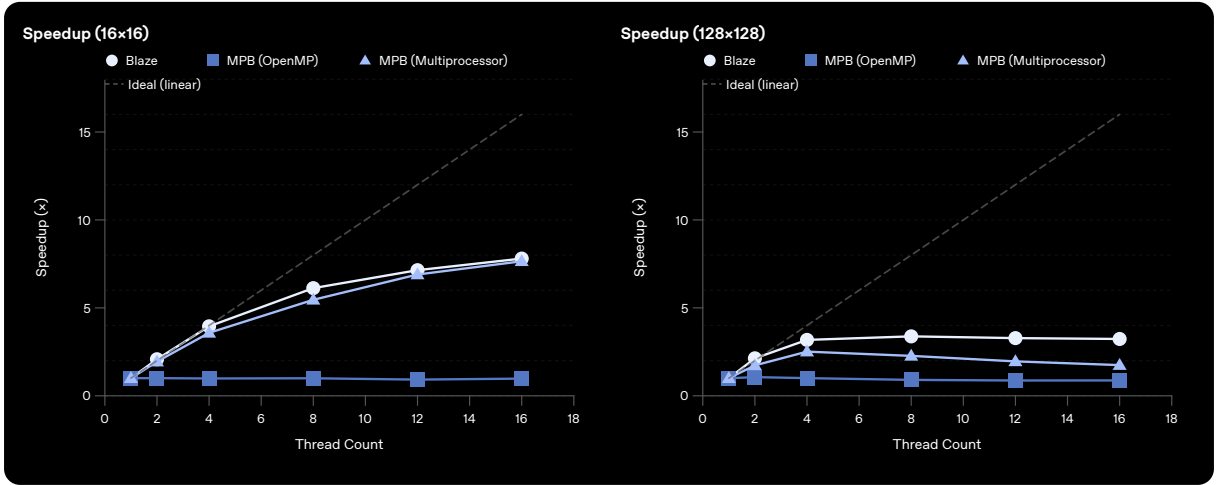


Figure 11: The same sweep expressed as speedup over the single-threaded result. Blaze tracks the ideal line closely at $N = 16$; MPB's threaded mode is flat in both panels.

8 Scaling with Resolution

Resolution sets the overall cost: the number of plane waves grows as N^2 and the FFTs as $N^2 \log N$. Figure 12 sweeps the grid from $N = 16$ to $N = 192$ and compares wall-clock time against MPB.

Blaze is faster at each sampled resolution in this dataset. At $N = 192$, the recorded times are approximately 8.0 s versus 39 s for TM and 12 s versus 70 s for TE. MPB timings vary non-monotonically with some grid sizes. FFT factorization and planning are possible contributors, requiring separate profiling.

The log-log plot compares empirical trends over the sampled resolutions. Its fitted slopes should not be extrapolated beyond this range without further measurements.

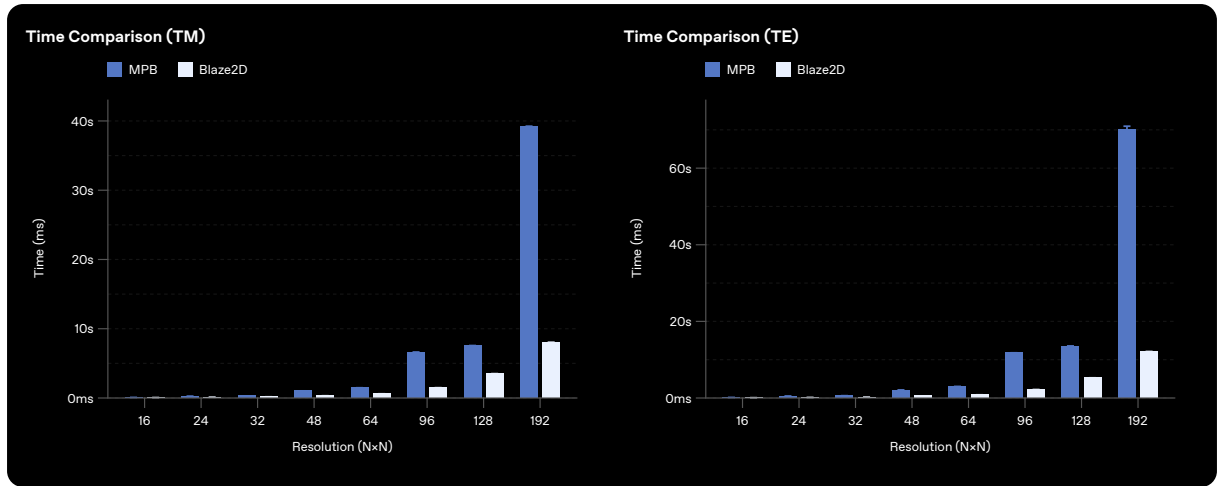


Figure 12: Wall-clock solve time against grid resolution for TM (left) and TE (right). MPB's cost is uneven across resolutions; Blaze's grows smoothly.

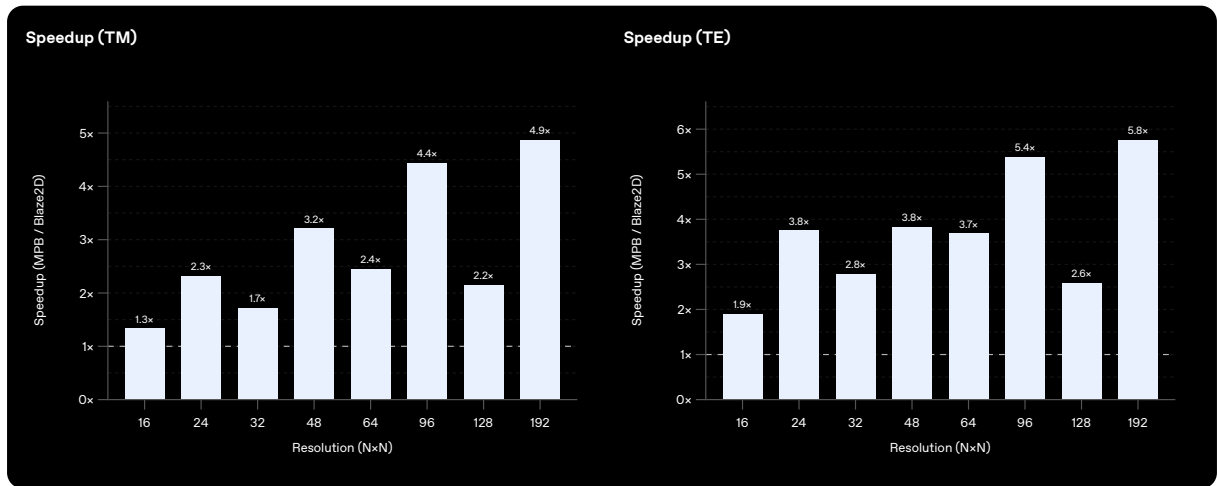


Figure 13: Blaze's speedup over MPB across the same sweep. The lead widens with resolution rather than saturating.

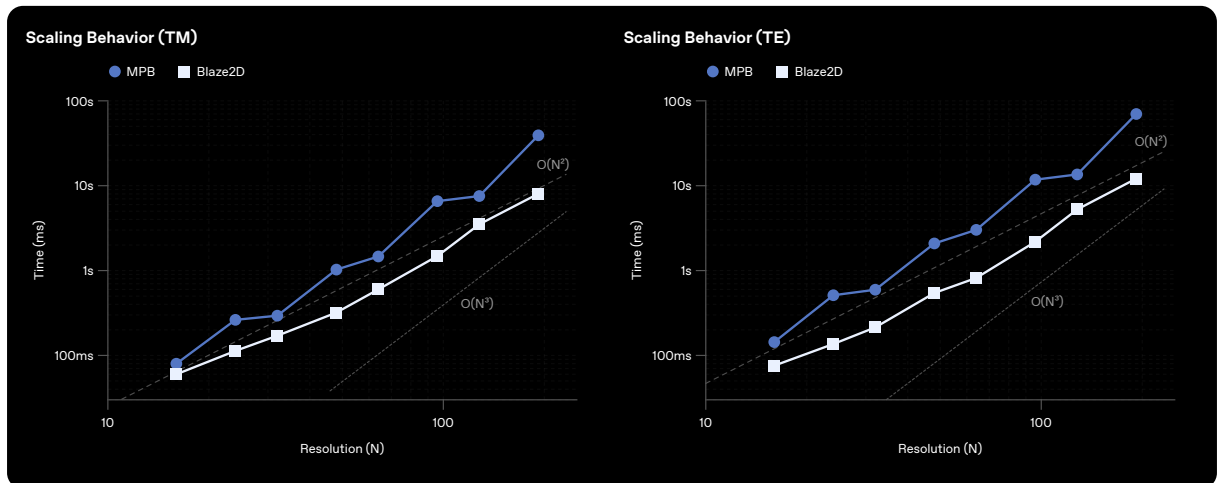


Figure 14: The same data on log-log axes with power-law fits, which makes the two solvers' scaling exponents directly comparable.

9 Convergence and Iteration Count

Beyond the cost of the *entire* single iteration, the number of iterations needed to converge each k -point matters.

Mean iteration counts are approximately 4.5 for Blaze TM and 5.6 for TE, compared with 7.5 and 11.9 for MPB. The recorded tolerances are $1e-4$ for Blaze and $1e-7$ for MPB. Both use eigenvalue changes in their stopping criteria; MPB documents this explicitly. The iteration counts therefore do not constitute an equal-tolerance comparison. Fresh residuals must be checked separately.

Maximum recorded iteration counts are 39 for MPB TM and 85 for MPB TE, compared with 12 and 24 for Blaze. Their locations vary along the path. The measurements do not isolate whether preconditioning, subspace updates, or stopping settings cause the difference.

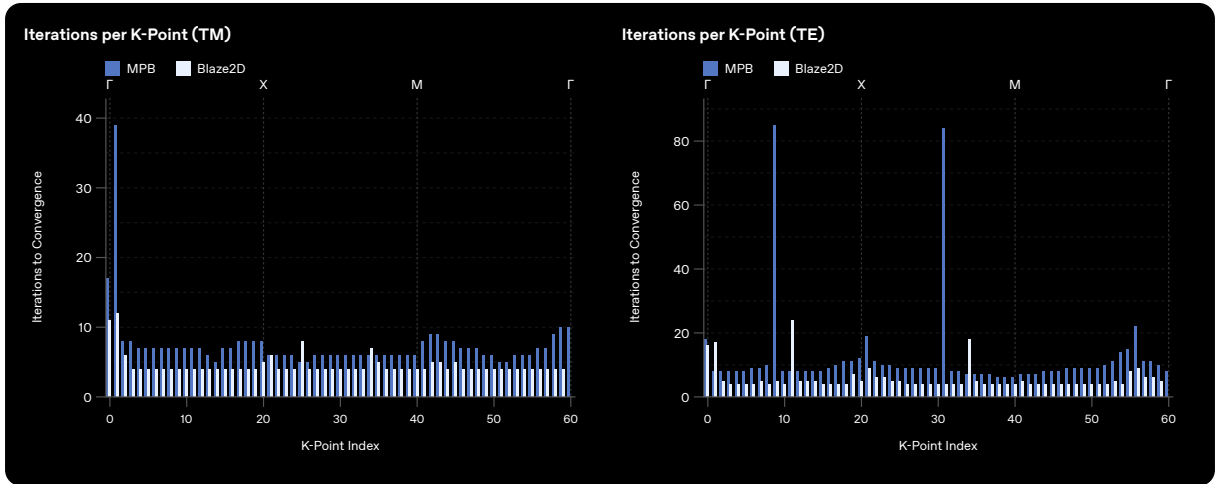


Figure 15: Mean iteration count per k -point for TM (left) and TE (right).

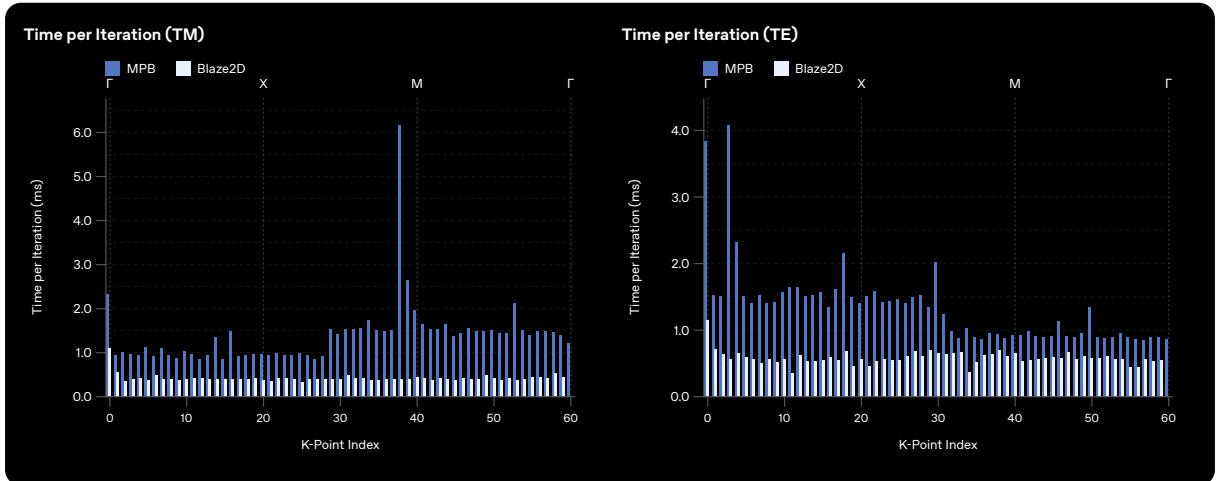


Figure 16: Time per iteration for the same runs. Blaze's per-iteration cost is lower as well, so the iteration-count advantage compounds.

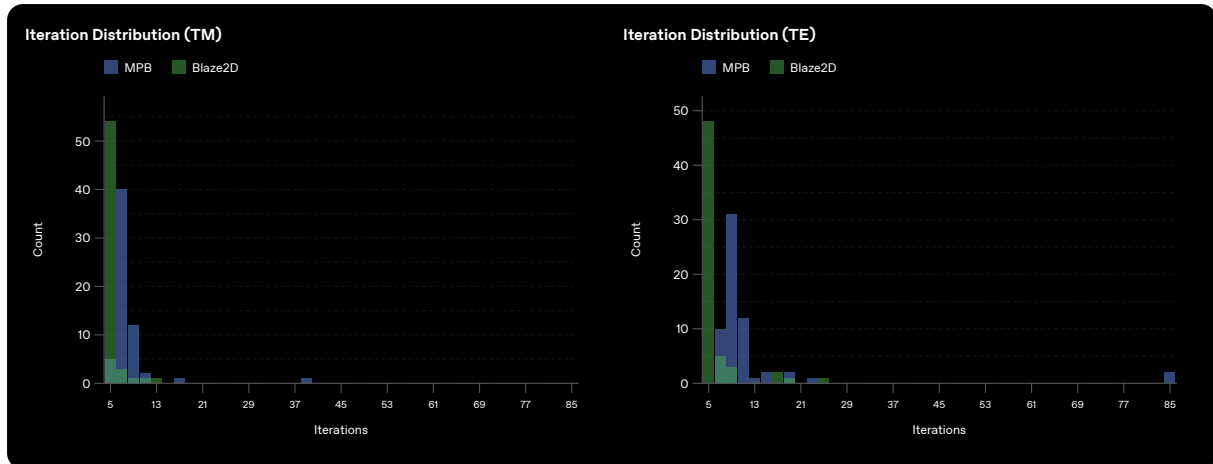


Figure 17: Distribution of iteration counts over all k -points. MPB's tail reaches 39 iterations for TM and 85 for TE; Blaze's stays short.

10 Dielectric Contrast

The dielectric-contrast sweep covers rod permittivities from 2 to 13. Blaze times vary little over these sampled configurations; MPB times vary more. This does not establish contrast-independent cost outside the measured range.

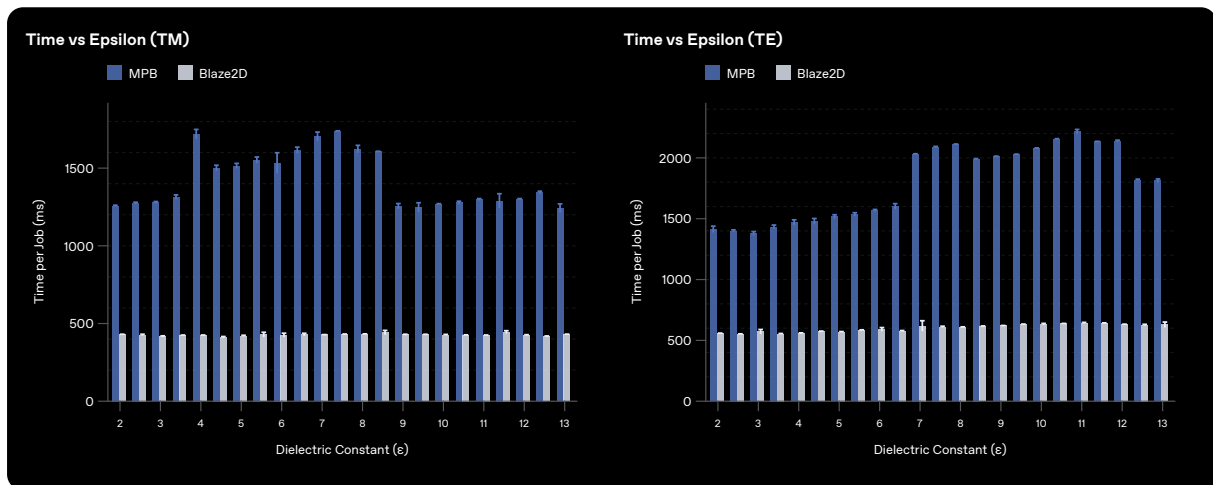


Figure 18: Solve time against rod permittivity for TM (left) and TE (right). Blaze is flat across the sweep; MPB is not.

11 Scaling with Band Count

The band-count sweep requests 4 to 20 bands. The recorded 20-band TM solve times are 1.9 s for Blaze and 3.9 s for MPB. Workspace size depends on band count even where process-memory measurements appear flat.

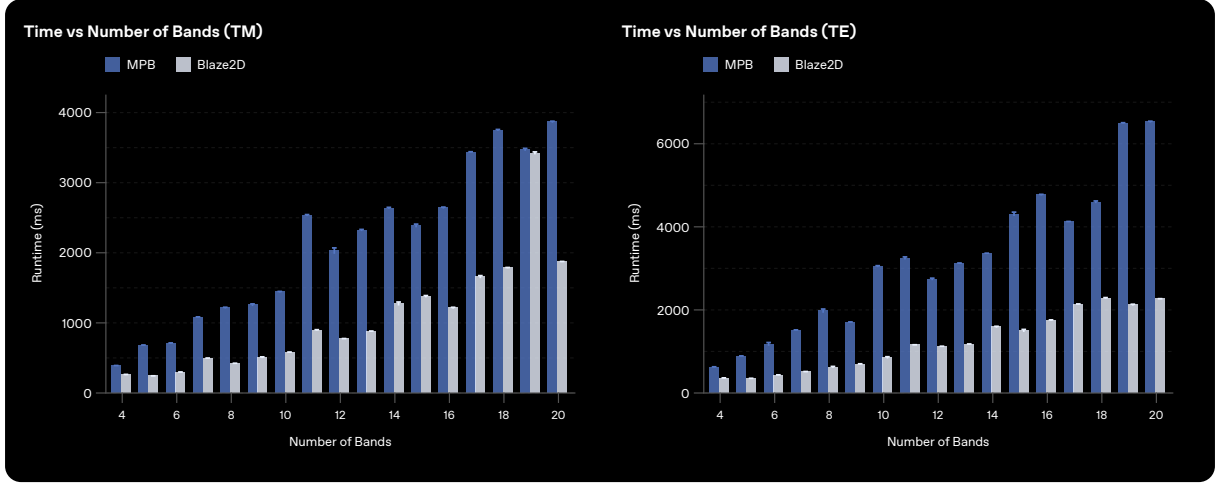


Figure 19: Solve time against the number of requested bands for TM (left) and TE (right).

12 Conclusion

The recorded experiments show lower Blaze wall-clock times and peak process memory for the tested configurations. The ratios depend on the stated numerical settings. A matched release comparison requires explicit hardware, solver revisions, retained outputs, and residual checks.

Blaze schedules independent configurations and uses mixed-precision storage with f64 critical reductions. It also exports projected operator and derivative data, including velocity matrices, effective masses, and formulation-specific geometric terms [8–10].

MPB exposes fields, eigenvectors, and group velocities. Blaze focuses on integrating parameter studies and projected extraction in one configuration and result contract.

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